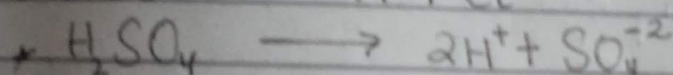
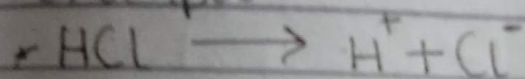


Q.2(i)

Arrhenius acids. An acid is a substance that ionizes in water to produce H^+ ions.

Example:-



(i)

HCl is a strong acid because it completely ionizes in water, producing a high concentration of H^+ ions.

(ii)

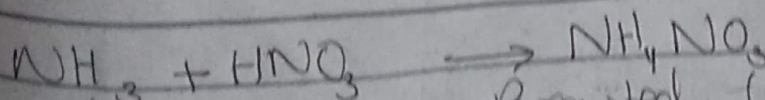
Ammonia is a weak base because it only partially ionizes in water to produce OH^- ions.

Q.3

(i) Arrhenius theory has its 2 limitations:-

- (1) This applies to aqueous solutions.
- (2) This does not explain why compounds like CO_2 & SO_2 are acidic. There is no H in CO_2 and SO_2 .

(ii)



According to the Bronsted Lowry theory:-

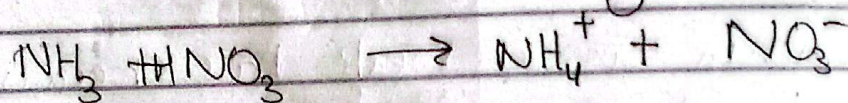
An acid is a proton donor.

A base is a proton acceptor.

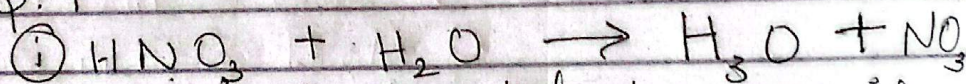
In this equation, HNO_3 is an acid because it donates H^+ protons to NH_3 . NH_3 is a base because it accepts H^+ protons from HNO_3 .

$$\text{NH}_3 + \text{HNO}_3 \rightarrow \text{NH}_4^+ + \text{NO}_3^-$$

In this equation, HNO_3 is an acid because it donates a proton (H^+) ions to NH_3 . NH_3 is a base because it accepts the H^+ proton from HNO_3 , forming NH_4^+ .



Q: 4

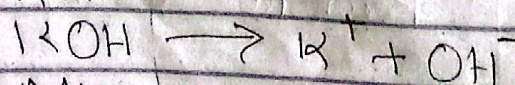


HNO_3 is the bronsted-lowry acid because it donates a proton (H^+) to H_2O .

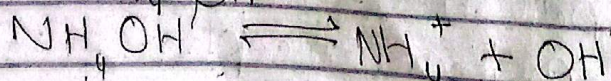
H_2O is the bronsted lowry base because it accepts a H^+ proton from HNO_3 .

(ii)

(a) KOH



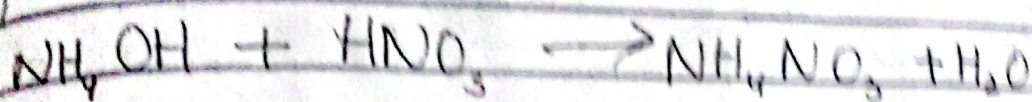
(b) NH_4OH



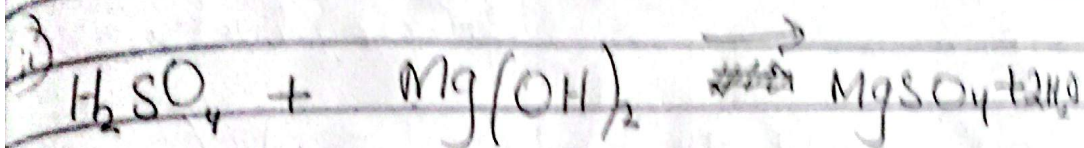
weak base :- NH_4OH , because it only partially

ionizes in water.

Q.5



Q.6



Q.6

Property	Acid	Base
Taste	Sour	Bitter
Effect on blue litmus	Turns red	no effect
Effect on red litmus	no effect	Turns blue
Effect on skin	Corrosive	Corrosive

Q.7

given: $\text{H}-\text{H} = 436 \text{ kJ/mol}$

$\text{I}-\text{I} = 151 \text{ kJ/mol}$

$\text{H}-\text{I} = 299 \text{ kJ/mol}$

ΔH° : Sum of bond dissociation energy of reactants — sum of formation energy of products.

$$\Delta H^\circ = [\text{BDE of H-H} + \text{BDE of I-I}] - (2 \times \text{BDE of H-I})$$

$$\Delta H^\circ = (436 + 151) - 2(299)$$

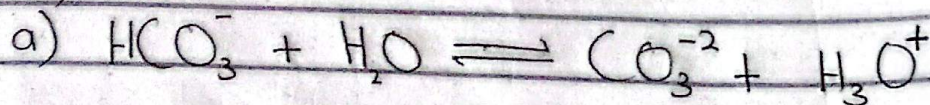
$$587 - 598$$

$$\Delta H^\circ = -11 \text{ kJ/mol} \quad \left\{ \begin{array}{l} \text{exothermic} \\ \text{reaction} \end{array} \right\}$$

Reason: The reaction is exothermic because its ΔH° is negative, indicating that energy is released.

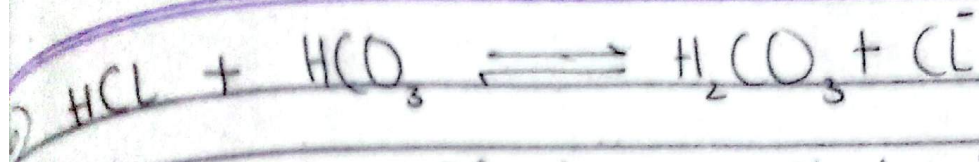
Q.8

i)



* HCO_3^- = an acid because it donates H^+ to H_2O .

* H_2O = a base because it accepts H^+ from HCO_3^- .



HCl = an acid because it donates H^+ to HCO_3^-

HCO_3^- = a base bcz it accepts H^+ from HCl.

(i) HCO_3^- is amphoteric bcz it can act as both an acid and a base.

* In (a) it acts as an acid by donating the protons.

* In (b) it acts as a base by accepting H^+ ~~from the~~

(iii) $\text{HS}^- + \text{H}_2\text{O} \rightleftharpoons \text{S} + \text{H}_3\text{O}^+$
Water is a proton acceptor (base) because it accepts H^+ from HS^- , forming H_3O^+ .

Q: 9

(i)

$$\begin{aligned}\text{mass of Zn} &= 6.5 \text{ g} \\ \text{molar mass} &= 65 \text{ g/mol} \\ \text{moles} &= \frac{6.5 \text{ g}}{65 \text{ g/mol}}\end{aligned}$$

$$= 0.1 \text{ mol}$$

(ii)



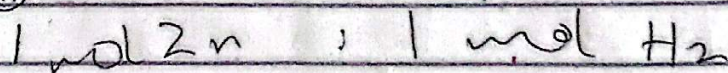
$$0.1 \times 2 = 0.2$$

molar mass of HCl :-

$$1 + 35.5 = 36.5 \text{ g/mol}$$

$$\text{mass} = 0.2 \times 36.5 = 7.3 \text{ g}$$

(iii)



$$0.1 \text{ mol} = \text{H}_2$$

molar mass of H₂ :-

$$2(1) = 2 \text{ g/mol}$$

$$\text{mass} = 0.1 \times 2$$

$$= 0.2 \text{ g}$$

(iv)

number of molecules

$$= \text{moles} \times N_A$$

$$0.1 \times 6.022 \times 10^{23}$$

$$6.022 \times 10^{22} \text{ molecules.}$$

(v)

$$\text{molar mass of ZnCl}_2 = 65 + 2(35.5) = 136 \text{ g/mol}$$

$$\text{moles of ZnCl}_2 = 0.1 \text{ mol}$$

$$\text{mass} = 0.1 \times 136 = 13.6 \text{ g}$$

(vi)

Therefore, 0.2 g of hydrogen gas and 13.6 g of zinc chloride are produced.

MCQs ::

1. C

2. A

3. C

4. D

5. B

6. D

7. A

8. B

9. B

10. A